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by [Mansour Saeed Mubarak Hamdan Al Hamdani](#) - Saturday, 30 August 2025, 8:35 AM

Agent Communication Languages

Languages of agent communication (ACLs) such as KQML (Knowledge Query and Manipulation Language) provide a structured language to support agent to agent communication within multi-agent systems. Their ability to standardize communication is one of the benefits, and it allows agents designed by other organizations to interact with each other without any problems (Labrou et al., 1999). KQML offers a protocol at a higher level, with performative verbs (e.g. ask, tell, achieve), which express the purpose of a message, making it clearer and less ambiguous. This standardization facilitates scalability and reusability which are important in complex systems required in the research of artificial intelligence (Finin et al., 1997). Also, the ontology-based framework of KQML enables agents to share and comprehend knowledge about domains, which enable them to collaborate in dynamic settings.

Nevertheless, KQML is limited. It is verbose and complex and this can add computational overhead and thus be inefficient than the simpler method invocation found in other languages such as Python or Java (Huhns and Singh, 1998). It needs a lot of professionalism to implement which may restrict its use. The other concern is security whereby open communication protocols can create vulnerabilities in systems unless they are securely guarded (Labrou et al., 1999). In addition, the strict format of KQML may fail to accommodate fast change of agent behaviour or changes in application requirements.

Otherwise, method invocation in Python or Java is a more direct, lightweight method. Python's dynamic nature and Java's object-oriented design allow for flexible, efficient agent interactions within a single application or framework (Lutz, 2013). These programming languages aid in quick prototyping and debugging, which are beneficial to real-life deployment.

In modern study, it is the selection of ACLs such as KQML and method invocation that is determined by the size and heterogeneity of the system (Wooldridge, 2009). KQML is best suited to large and varied networks of agents whereas Python and Java are suited to controlled and integrated environments. Knowledge of such trade-offs is vital to implementing intelligent systems at their best, and in line with both the learning outcomes of agent models and modern research challenges.

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